IT2140 - Database Design and Development

Year 2, Semester 1

Tutorial 07

Learning outcomes:

- Identify candidate keys using functional dependencies.
- Analyze the normal form of a relation.
- Detect violations and normalize relational schemas.

1. Consider the following functional dependencies for a relation R: (A, B, C, D, E, F)
   F= {A→C, C→D, D→B, E→F}

   a. Find all keys of R.

2. Consider the following functional dependencies for a relation: R (A, B,C ,D , E, F)
   F = {AB→C , D C →A E , E →F}

   b. Find all the keys of R.

3. Consider a relation R= (A,B,C,D,) with the following functional dependencies:
   F= {CE →D,D → B,C → A }

   a. Find all candidate keys in R
   b. Which normal for is R in? c. If the relation is not in BCNF, convert it to a set of relations in BCNF through decomposition.

4. Consider a relation R (A, B, C, D, E), with the following set of functional dependencies over R: F = {A → BC, BC → E, E → DA }

   a. Find all the keys in R.
   b. Is R in BCNF? If R is not in BCNF, convert it to a set of BCNF relations.

5. Consider the following functional dependencies for a relation R(A,B,C,D,E)
   F = {AB→C , A B →D, D→A , B C →D , B C →E}

   a. Find all the keys of R.
   b. Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations.

6. Consider a relation R=(A,B,C,D,E) with the following functional dependencies: F= {BC→ADE, D→B}
    a. Find all candidate keys in R
    b. Which normal for is R in?
    c. If the relation is not in BCNF, convert it to a set of relations in BCNF through decomposition